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Notifications

15. Properties of the matrix exponential

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Some properties and computation



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want to give it explicitly.

Start with any fundamental matrix calculated--

you should forgive the expression--
using eigenvalues and eigenvectors
and putting the solutions into the columns.

Evaluate it at 0, take its inverse,

multiply the two, and what you end up with

has to be the same as the thing calculated

with that infinite series.

Properties of the matrix exponential

1. $e^{\mathbf{0}} = \mathbf{I}$ (here $\mathbf{0}$ is the zero matrix)
Proof: $e^{\mathbf{0}} = \mathbf{I} + \mathbf{0} + \frac{\mathbf{0}^2}{2!} + \dots = \mathbf{I}$

2. $\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t}$
Proof: Take the derivative of $e^{\mathbf{A}t}$ term by term.

3. If $\mathbf{AB} = \mathbf{BA}$, then $e^{\mathbf{A+B}} = e^{\mathbf{A}}e^{\mathbf{B}}$. Here are a few (but not all) special cases when $\mathbf{AB} = \mathbf{BA}$:

- $\mathbf{A} = c\mathbf{I}$
- $\mathbf{B} = -\mathbf{A}$
- $\mathbf{B} = \mathbf{A}^{-1}$

Warning: In general, this fails, that is, if $\mathbf{AB} \neq \mathbf{BA}$, then usually $e^{\mathbf{A+B}} \neq e^{\mathbf{A}}e^{\mathbf{B}}$, since there are counterexamples even for real 2×2 matrices.

4. If $\mathbf{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, then $e^{\mathbf{A}} = \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix}$.
Proof: $\mathbf{A}^2 = \begin{pmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{pmatrix}$, $\mathbf{A}^3 = \begin{pmatrix} \lambda_1^3 & 0 \\ 0 & \lambda_2^3 \end{pmatrix}$, and so on. Thus

$$e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \dots = \begin{pmatrix} 1 + \lambda_1 + \frac{\lambda_1^2}{2!} + \dots & 0 \\ 0 & 1 + \lambda_2 + \frac{\lambda_2^2}{2!} + \dots \end{pmatrix} = \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix}.$$

(A similar statement holds for diagonal matrices of any size.)

5. Given $\mathbf{A}$ is **diagonalizable**, that is,

$$\mathbf{A} = \mathbf{SDS}^{-1}, \tag{4.173}$$

where $\mathbf{D}$ is the diagonal matrix of eigenvalues and $\mathbf{S}$ is the invertible matrix whose columns are eigenvectors:

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \end{pmatrix} \tag{4.174}$$

$$\mathbf{S} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{pmatrix} \quad \text{where } \mathbf{v}_i \text{ corresponds to } \lambda_i. \quad (4.175)$$

Then

$$e^{\mathbf{A}} = \mathbf{S}e^{\mathbf{D}}\mathbf{S}^{-1}. \quad (4.176)$$

Proof: Expand $e^{\mathbf{A}}$ as a power series and use the following cancellations for each term:

$$\mathbf{A}^n = \mathbf{S}\underbrace{\mathbf{D}\mathbf{S}^{-1}\mathbf{S}}_{\text{cancels}}\cdots\underbrace{\mathbf{S}^{-1}\mathbf{S}\mathbf{D}\mathbf{S}^{-1}\mathbf{S}}_{\text{cancels}}\cdots\underbrace{\mathbf{S}^{-1}\mathbf{S}\mathbf{D}\mathbf{S}^{-1}}_{\text{cancels}} = \mathbf{S}\mathbf{D}^n\mathbf{S}^{-1}. \quad (4.177)$$

6. $e^{\mathbf{A}t} = \mathbf{X}(t)\mathbf{X}(0)^{-1}$ for any fundamental matrix $\mathbf{X}$.

Proof: Both $e^{\mathbf{A}t}$ and $\mathbf{X}(t)\mathbf{X}(0)^{-1}$ satisfy the matrix differential equation $\dot{\mathbf{Y}} = \mathbf{A}\mathbf{Y}$ and the same initial conditions $\mathbf{Y}(0) = \mathbf{I}$. The uniqueness and existence theorem then guarantees these to be the same. (You can also think of the columns of $e^{\mathbf{A}t}$ separately and apply the existence and uniqueness theorem.)

Problem 15.1 Use the matrix exponential to find the solution to the system

$$\begin{aligned} \dot{x} &= 2x + y \\ \dot{y} &= 2y \end{aligned}$$

satisfying $x(0) = 5$ and $y(0) = 7$.

Solution: This is $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{A} := \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Write $\mathbf{A} = \mathbf{D} + \mathbf{N}$ with $\mathbf{D} := \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (diagonal) and $\mathbf{N} := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $\mathbf{N}^2 = \mathbf{0}$, so

$$e^{\mathbf{N}t} = \mathbf{I} + \mathbf{N}t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Then $\mathbf{D}t$ and $\mathbf{N}t$ commute (a scalar times $\mathbf{I}$ commutes with any matrix of the same size), so

$$\begin{aligned} e^{\mathbf{A}t} &= e^{\mathbf{D}t + \mathbf{N}t} \\ &= e^{\mathbf{D}t}e^{\mathbf{N}t} \\ &= \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{pmatrix} \\ \begin{pmatrix} x \\ y \end{pmatrix} &= e^{\mathbf{A}t} \begin{pmatrix} 5 \\ 7 \end{pmatrix} \\ &= \begin{pmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} 5 \\ 7 \end{pmatrix} \\ &= \begin{pmatrix} 5e^{2t} + 7te^{2t} \\ 7e^{2t} \end{pmatrix}. \end{aligned}$$

Remark: The matrix exponential is a fundamental matrix even when $\mathbf{A}$ is deficient, as in the example above.

Compute the exponential matrix

1/1 point (graded)

Compute the exponential matrix $e^{\mathbf{A}}$ for $\mathbf{A} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$.

(Enter `[[a,b],[c,d]]` for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

$e^{\mathbf{A}} =$

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[[0.20883325, 0.15904619], [0.15904619, 0.20883325]]



Answer: [[cosh(1)/e^2, sinh(1)/e^2],[sinh(1)/e^2,cosh(1)/e^2]]

Solution:

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$$(-2\mathbf{I})\mathbf{B} = \mathbf{B}(-2\mathbf{I}). \tag{4.179}$$

This implies that $e^{\mathbf{A}} = e^{-2\mathbf{I}}e^{\mathbf{B}}$. Let us compute $e^{\mathbf{B}}$:

$$e^{\mathbf{B}} = \mathbf{I} + \mathbf{B} + \frac{\mathbf{B}^2}{2} + \frac{\mathbf{B}^3}{3!} + \frac{\mathbf{B}^4}{4!} + \cdots \qquad \left(\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) \tag{4.180}$$

$$= \mathbf{I} + \mathbf{B} + \frac{\mathbf{I}}{2} + \frac{\mathbf{B}}{3!} + \frac{\mathbf{I}}{4!} + \cdots \qquad \text{since } \mathbf{B}^2 = \mathbf{I} \tag{4.181}$$

$$= \mathbf{I} \left(1 + \frac{1}{2} + \frac{1}{4!} + \cdots\right) + \mathbf{B} \left(1 + \frac{1}{3!} + \frac{1}{5!} + \cdots\right) \tag{4.182}$$

$$= \mathbf{I} \cosh(1) + \mathbf{B} \sinh(1) = \begin{pmatrix} \cosh(1) & \sinh(1) \\ \sinh(1) & \cosh(1) \end{pmatrix}. \tag{4.183}$$

Therefore,

$$e^{\mathbf{A}t} = e^{-2t\mathbf{I}} \begin{pmatrix} \cosh(1) & \sinh(1) \\ \sinh(1) & \cosh(1) \end{pmatrix} = e^{-2} \begin{pmatrix} \cosh(1) & \sinh(1) \\ \sinh(1) & \cosh(1) \end{pmatrix}. \tag{4.184}$$

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15. Properties of the matrix exponential

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